Module 7 - Slide Deck Video Transcription

1 Hi, I'm Jordan Reyes from Tallgrass Advisory, a fictional character presented by an AI avatar, on purpose, because this course is about using AI well and being open about it. Welcome to your fourth rotation: Tax, working for Grace Lindqvist, the Velox Director of Tax. This week the IRS has asked Velox a question, and you'll have AI draft the research, check its own sources against documents you provide, and calculate what's at stake. Then you'll check two results yourself, because the title of this week is Show Me the Source. A source nobody has opened is not a source.

2 Here's Grace's situation. The IRS sent Velox a letter about its 2025 tax return. It lists nine pieces of equipment at the South Carolina plant, a total cost of about forty point four million dollars, and notes that Velox deducted a hundred percent of that cost in the first year. The IRS wants three things: the date each purchase contract was signed, the date each item was placed in service, and an explanation of why a hundred percent applies, including the legal authority. The deadline is November ninth. Grace decides what Velox sends, and Grace signs it. You prepare the work.

3 First, a little tax vocabulary, kept short on purpose. Tax professionals answer a question like this in a research memo with five parts: the question, the facts, the rules, the analysis, and the conclusion. The rules come from authorities. An authority is a source anyone can open and read: the Internal Revenue Code, which is the federal tax law, Treasury regulations, IRS notices, and court decisions. A citation is the address of an authority, like section one sixty-eight, or IRS Notice twenty twenty-six eleven. A friend's memory, a blog post, or what an AI says from memory is not an authority.

4 Now the problem with AI. It writes text that sounds right. With citations, three things can go wrong. It can cite a real rule with the wrong section number. It can cite a rule that has since changed. Or it can cite something that doesn't exist. Often, an AI memo is mostly right, and you still can't rely on it, because you can't tell which parts are right without opening the sources. Asking, are you sure, doesn't fix that, because AI sounds sure either way. What helps is to give the AI the sources, make it quote them, and then open the ones that matter yourself.

5 Let me show you with a made-up example, so I don't give away your assignment. Cedar Valley Brewing Co. got a similar IRS letter about four pieces of equipment, four million twenty-five thousand dollars in all, deducted at a hundred percent. I gave Claude the letter and the equipment schedule, with no sources. My request: a short research memo, then a

table of every source it cited, with the exact words relied on. No template, no rules. This is what a rushed analyst might do.

6 Here is Claude's real output. The conclusion was good: the hundred percent claim holds only for the bottling line. The tanks and racks were contracted before January twentieth, twenty twenty-five, so they get forty percent. The lab equipment has no contract date. Net over-deduction: about one point three three million. Then it listed five citations, and said: I drafted this from memory with no source access. Verify each citation. It was honest, and still you can't send this, because nobody can tell which citations are reliable.

7 So here is the fix, and it has a name: source-grounded prompting. Practitioners also call it grounding or open-book prompting, and it's the idea behind retrieval-augmented generation, or R-A-G. You attach the sources, tell the AI to use only them, to label and quote the exact words for every rule, and to write, not in the sources, when they don't answer. Why it helps: you can check every statement against text you can see, and a gap shows up as a flag instead of a guess. I attached a short Source Packet, notes S one to S eight. The first six are word-for-word excerpts from an IRS notice and an IRS publication, not summaries.

8 This is Claude's cite-check, condensed. Three citations were marked Not in the packet: the Code subsections, the section one seventy-nine limits, and the regular depreciation subsections, because the packet quotes IRS guidance, not the Code itself. One was Supported, with a difference. Two were marked Different. The forty percent belongs to the earlier law. And the regulation: Claude had said the acquisition date is the date the contract is signed, but the packet says it is the later of four dates, including the end of any cancellation period. Notice that Not in the packet is a flag, not a verdict. A person opens the original, or leaves it out.

9 Then Claude redid the memo from the packet only, quoting its labels and listing its assumptions. It wrote not in the sources for section one seventy-nine, state tax, interest, and penalties, and withdrew the section one seventy-nine point. Then it used code and showed each item. The tanks were over-claimed by one million seventy-nine thousand nine hundred forty-six dollars, the racks by two hundred forty-six thousand eight hundred forty-five, the bottling line by zero. For the lab equipment it showed both outcomes and did not guess the missing date. Total: between one point three three and one point three eight million. Extra federal tax at twenty-one percent: about two hundred seventy-nine to two hundred eighty-nine thousand. And one assumption matters: Claude treated each signing date as the acquisition date, though a cancellation period could make it later.

10 What to notice. One: the first memo was good, and still couldn't be used. Two: given the packet, Claude sorted its own citations, caught that the acquisition date is not simply the signing date, and withdrew a point the packet didn't support. Three: it used code and gave both outcomes for the missing date. Four: the answer is only as good as the packet. Five: Claude's cite-check is a draft too. AI checking its own work can be wrong, so a person opens the sources that matter. That's Part Five: you check two findings yourself, with your own arithmetic or your own reading of a source.

11 Now the memo you write for Grace. Start with the most important sentence: what you found, how large it is, and that it rests on the Source Packet only. Then the rules, with packet labels, the numbers that matter, what is missing or not in the sources, and the next step for a person. Three hundred words at most. Roles and asset I-D codes. Don't blame anyone for the missing date. And this week's reusable piece is your Research Memo Template.

12 This week's ethics checkpoint is area two of the Velox Responsible AI Policy, accuracy: verify facts, figures, and sources before you rely on them. Because the response goes out under a signature, area five, human accountability, applies too. AI may draft, compare, and calculate. It must not be the only check on its own citations, present a source as checked when no person has opened it, give a number without showing how it was built, or send or sign anything to the IRS. A named person, here the Director of Tax, opens the sources, checks the arithmetic, and signs.

13 Here's your week. Try the letter in a plain chat, then learn the ideas and judge five AI suggestions. Write your Research Memo Template, ask Claude for a memo with no sources, then give it the Source Packet and have it check its citations, redo the memo, and calculate with code. If your plan can search the web, ask it to look up the IRS originals too. Check two findings yourself, write Grace a short memo, and choose a lens from six equal options. Then get feedback, revise once, take the quiz, and reflect. Show me the source. I'll see you in the module.